

Spiral Bevel Gear

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Gear terminology

Basic Tooth System Parameters

- **Module (m)** - tooth size, mm of pitch diameter per tooth. Determines overall tooth size and combined with tooth count, sets pitch diameter
- **Pressure angle (φ)** - 20° is the modern AGMA/Gleason standard for spiral bevel (older systems used 14.5°). Controls load angle, tooth shape and undercutting resistance
- **Number of teeth (N)** - separately for pinion (Np) and gear (Ng). Their ratio gives you the gear ratio Ng/Np.
- **Circular pitch** - tooth spacing measured along the pitch circle

Bevel Specific Geometry (Differ for spur / helical)

- **Shaft angle (Σ)** - Angle between the 2 shaft axes; 90° in central to blade layout, but bevel theory generalizes to any angle.

- **Pitch cone angle (γ)** - the half angle of the pitch cone for pinion and gear separately; this is the bevel gear equivalent for the pitch diameter for spur gears and its what makes each tooth a different size along its length
- **Cone distance (outer, mean, inner)** - the slant length from the pitch apex to the outer edge (heel), the middle of the face width, or the inner edge (toe) of the tooth. Because tooth size varies along the face, all bevel calculations are typically done at the mean cone distance, not the outer edge
- **Face width (F)** - axial length of the tooth along the cone. as a rule of thumb kept to roughly a third of the outer cone distance (going longer does not help much the tooth gets so small near the toe that it contributes little strength , but it add cost and weight)
- **Pitch apex** - the reference point where the two pitch cones axes theoretically intersect, and the axial distance from that apex to the back face of the gear. Mounting distance is the single most safety critical dimension for assembly.
- **Back cone** - used to calculate strength analysis using Lewis Equation
- **Addendum / Dedendum** - Deliberately unequal between pinion and gear to balance sliding wear as the ratio increases.

Spiral Specific Terminology

- **Spiral angle (ψ)** - the angle of the tooths curved trace relative it the cone element; 35 is the AGMA/Gleason standard default. This is the parameter that gives spiral bevel its gradual engagement and higher contact ratio (directly analogous to helix angle in helical spur gear)
- **Hand of spiral (left-hand / right-hand)** - determines the direction of the axial trust generated at the mesh for a given direction of rotation. Getting this wrong will result in the thrust goes the wrong way into your bearing.
- **Cutter radius** - spiral bevel teeth are generated by a circular cutter, cutter radius affects the actual curvature of the tooth trace and interacts with the spiral angle.
- **Lengthwise curvature / tooth crowning** - the deliberate convexity along the tooth length that jeeps contact centered even with small mounting errors

Load Terminology

- **Tangential force** -transmits the actual torque
- **Radial force** - pushes the 2 gears directly apart from the pressure angle
- **Axial thrust force** - unique to bevel geometry, generated by the cone itself. This is the force your thrust bearings must be sized against
- All the three act simultaneously at the mesh point and must each be resolved into the bearing / housing design

Simple Explanation

- **Pinion** - The smaller gear of a pair in contact
- **Gear** - The other one
- **Pitch diameter** - diameter of the pitch circle. Is the basic diameter of the pinion and gear which when divided by each other equals the ratio.
- **Module** - Pitch diameter divided by number of teeth
- **Circular pitch** - Tooth to tooth distance on the pitch circle or tooth thickness + width of the space p = pi()d/N
- **Addendum** - Distance from the pitch circle to the addendum circle = 1/P in.
- **Dedendum -** Distance from the pitch circle to dedendum circle = 1.25/P in.
- Pitch Circle - Theoretical circle upon which all calculation is based
- Circular pitch - The distance from one teeth to the next, along the pitch circle $p = \pi d/N$
- Module = d/N, pitch circle / number of teeth $p = \pi m$
- Diametral Pitch = $P = N/d$, $pP = pi$
- **Whole Depth -** or total depth of a gear tooth equals the addendum plus the dedendum. The whole depth equals 2.250 divided by the diametral pitch
- **Working Depth** - of a tooth equals the whole depth minus the height of the radius at the base of the tooth. The working depth equals 2.000 divided by the diametral pitch.

- **Clearance -** equals the whole depth minus the working depth. The clearance equals the height of the radius at the base of the tooth.

Formulas

Contact Ratio

$$CR = \frac{\sqrt{r_{ap}^2 - r_{bp}^2} + \sqrt{r_{ag}^2 - r_{bg}^2} - c \sin \phi}{p_b}$$

- a - addendum circle
- b- base circle
- pb - base pitch = $\pi d_b / N$
- db - $d \cos \phi$ & pb - $p \cos \phi$
- r_ap, r_ag - addendum radii of the mating pinion and gear
- r_bp, r_bg - base circle radii of the mating pinion and gear

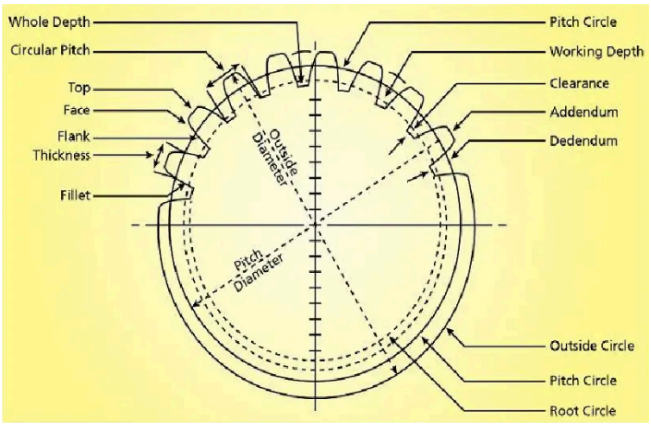
Greater the contact ratio, smoother and quieter

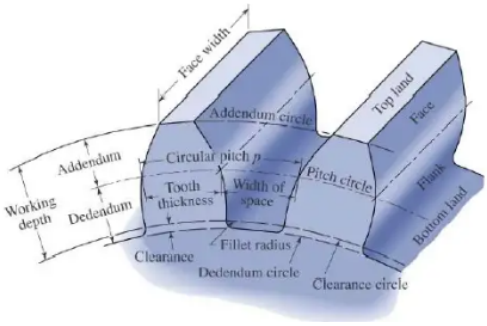
CR = 2 it means at least 2 pairs of teeth are theoretically in contact

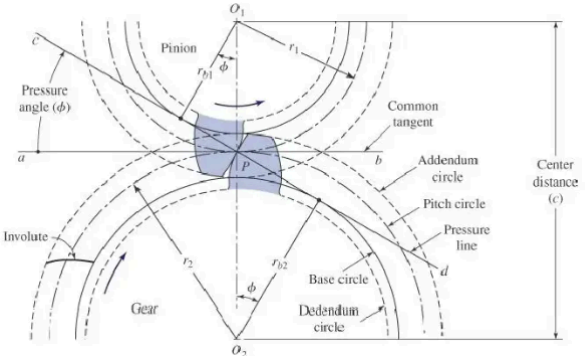
Standard Gear Teeth

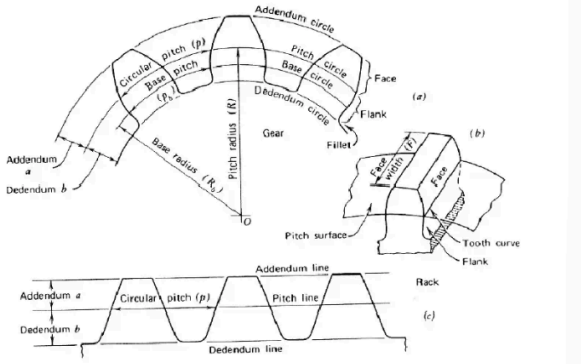
Item	20° full depth	20° Stub	25° full depth
Addendum a	1/P	0.8/P	1/P
Dedendum	1.25/P	1/P	1.25/P
Clearance f	0.25/P	0.2/P	0.25/P
Working depth	2/P	1.6/P	2/P
Whole depth	2.25/P	1.8/P	2.25/P
Tooth Thickness	1.571/P	1.571/P	1.571/P
Face width	9/P < b < 13/P	9/P <b < 13/P	9/P <b < 13/P

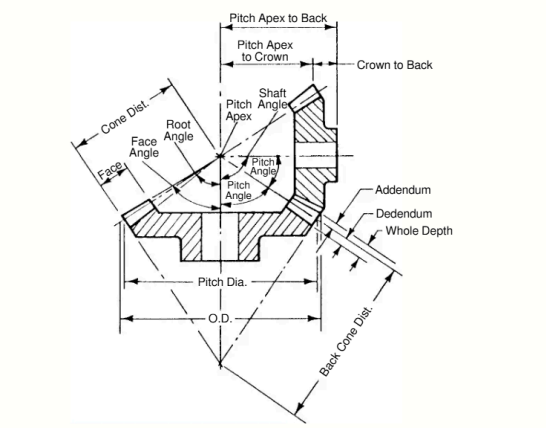
Figures

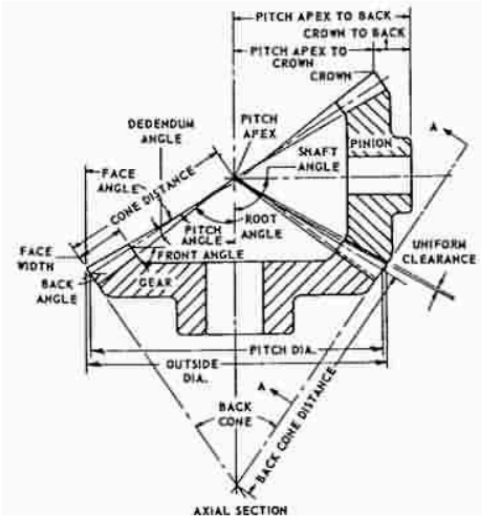

Whole Depth
Circular Pitch
Top
Face
Flank
Thickness
Fillet
Outside Diameter
Pitch Diameter
Pitch Circle
Working Depth
Clearance
Addendum
Dedendum
Outside Circle
Pitch Circle
Root Circle

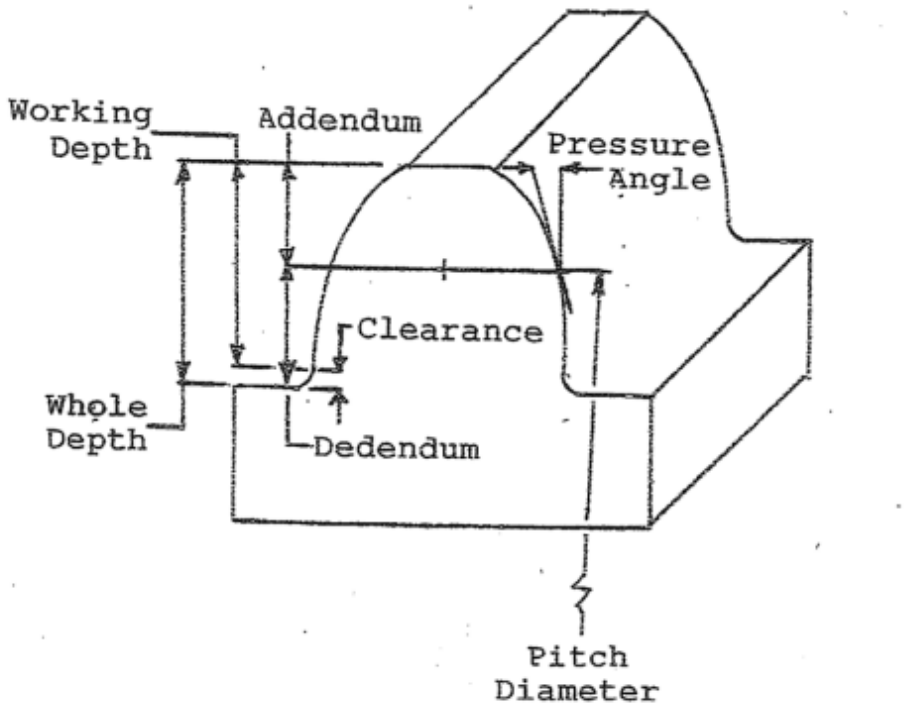

Face width
Addendum circle
Top land
Face
Addendum
Circular pitch p
Working depth
Dedendum
Tooth thickness
Width of space
Pitch circle
Flank
Bottom land
Fillet radius
Clearance
Dedendum circle
Clearance circle


Pinion
Pressure angle (φ)
Common tangent
Addendum circle
Center distance (c)
Pitch circle
Pressure line
Involute
Base circle
Gear
Dedendum circle


Addendum circle
Circular pitch (p)
Base pitch
Pitch circle
Base circle
Face
Flank
Fillet
Dedendum circle
Gear
Addendum a
Dedendum b
Base radius (Rb)
Pitch radius (R)
Face width (F)
Pitch surface
Tooth curve
Addendum line
Rack
Circular pitch (p)
Pitch line
Dedendum line


Pitch Apex to Back
Pitch Apex to Crown
Crown to Back
Cone Dist.
Shaft Angle
Pitch Apex
Root Angle
Face Angle
Face
Pitch Angle
Pitch Angle
Addendum
Dedendum
Whole Depth
Pitch Dia.
O.D.
Back Cone Dist.


PITCH APEX TO BACK
CROWN TO BACK
PITCH APEX TO CROWN
CROWN
DEDENDUM ANGLE
PITCH APEX
PINION
A
SHAFT ANGLE
FACE ANGLE
CONE DISTANCE
ROOT ANGLE
PITCH ANGLE
FRONT ANGLE
FACE WIDTH
BACK ANGLE
GEAR
UNIFORM CLEARANCE
PITCH DIA.
OUTSIDE DIA.
A
BACK CONE
BACK CONE DISTANCE
AXIAL SECTION



Forces

There are 3 fundamental forces that the bevel gear tooth experience.

Force component	Primary direction of action
Tangential (Ft)	Acts along the tangent to the pitch circle
Radial (Fr)	Acts toward the center of the gear
Axial (Fa)	Acts along the gear’s shaft axis

- Tangential Force (Ft)
  - This is the useful component. The tangential force is what actually transmits the torque and the power. It’s directly proportional to the torque being applied to the gear
- Radial Force (Fr)
  - The pressure angle of the gear creates a separating force. The radial component is the part of this forces that pushes the 2 gears directly away from each other, perpendicular to their shafts.

- Axial Force (Fa)
    - The cone angle of the bevel gears also generates a thrusting force. This axial force pushes each gear along its shaft axis. This is critical factor that differentiates bevel gears from simple spur gears

Design Element	Key forces to Consider	Why It is Critical
Bearing Selection	Radial & Axial	Tapered roller bearings are often needed to handle the combined loads
Shaft Deflection	Tangential & Radial	The shaft must be stiff enough to resist bending and maintain gear alignment
Housing Design	All Three	The housing must securely support the bearing and prevent misalignment

Bevel Gear Types and Use Case

Straight bevel gears

- When speeds are less than 300 m/min - at higher speeds, straight bevel gears may be noisy
- When factors are light, or for high static loads when surface wear is not a critical factor
- When space, gear weight, and mountings are a premium. This includes planetary gear sets, where space does not permit the inclusion of rolling - element bearings

Coniflex gears

- Tolerance for small amounts of misalignment in the assembly of the gears and some displacement of the gears under load without concentrating the tooth contact at the ends of the teeth
- Coniflex gears are capable of transmitting larger loads than the straight bevel gears

Spiral bevel gears

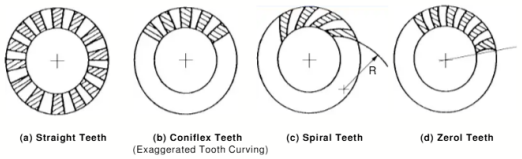
- Have oblique teeth which contact each other gradually and smoothly from one end to the other
- Have two or three teeth in contact at all times.
    - This mean the load is distributed over 2 - 3 tooth pairs instead of one so each tooth caries less stress. Resulting in spiral bevel gear transmit more torque power for the same size compared to straight bevel gears
    - Smoother and quieter operation - because engagement and disengagement happen gradually and overlap, there is no sudden tooth to tooth impact. This reduces vibration and noise. This why it is used in high speed application
    - Gradual engagement - the angled tooth contact starts at one point and progressively sweeps across the face width as the gears rotate, rather than the full tooth face engaging all at once
- The overlapping tooth action transmits motion more smoothly and quietly than with straight bevel gears

Zerol bevel gears

- Have similar curved teeth similar to those of the spiral bevels, but with zero spiral angle at the middle of the face width
- Zerol can be used in the place of straight bevel without mounting changes
- Used where ground tooth precision gears are generally required
- Zerol gears have curved teeth but zero spiral angle. This gives them the gradual tooth engagement. This means they run smoother and quieter than straight bevel gears.
- Zerol gears are recommended for high-speed, high-load applications where reversing direction is necessary.

Hypoid gears

Hypoid gears are designed for shafts that are offset. This means their axes does not intersect. This offset is their defining feature.



Contact Ratio

Contact ratio is the average number of teeth in contact at any one time. The higher the ratio the better.

More teeth sharing the load means smoother power transmission. This reduces vibration and noise significantly.

Contact Ratio	Operational Effect	Benefit
Low (< 1.2)	Rough, Noisy	Lower cost
High (> 1.2)	Smooth, Quiet	Increased Durability

A higher contract ratio means the load is distributed across multiple teeth pairs. One pair of teeth is already in full contact before the preceding pair disengages.

The overlap is the key. It prevents abrupt load transfers. Abrupt transfers are a major source of noise and impact stress in gear systems.

How Contact Ratio Reduces Wear

With the load shared the peak stress on any single tooth is much lower. This reduces the risk of pitting scoring and eventual tooth failure. There is less instantaneous pressure, which also minimizes heat generation and material fatigue over millions of cycles.

Feature	Low Contact Ratio	High Contact Ratio
Load Distribution	Concentrated on one pair	Shared across multiple pairs
Noise Level	Higher	Lower
Vibration	Significant	Minimal

Wear Rate	Faster	Slower
Gear Lifespan	Shorter	Longer

Pressure Angle in Bevel Gear System

The pressure angle is the angle between the line of force and the line tangent to the pitch circles at the point of contact

- Line of force is the direction of the force exerted by the driving tooth in the driven tooth
- Tangent line is a line tangent to both pitch circles at the pitch point
- Pressure angle is the angle between these 2 lines

A larger pressure angle also increases the radial load on bearings. This force pushes the gears apart. The system's bearing and housing must be strong enough to handle this increased load without deflection.

Undercutting

Undercutting is a manufacturing issue. It happens when designing gears with a low tooth count and a small pressure angle. The cutting tool can remove material from the base of the tooth, weakening it severely.

Pressure Angle	Tooth Strength	Bearing Load	Undercutting Risk (with low tooth count)
14.5°	Lower	Lower	High
20°	Standard	Standard	Moderate
25°	Higher	Higher	Low

Comparison between Gear Types

Bevel Gear Vs Spur Gear

Feature	Spur Gear	Bevel Gear
Base Shape	Cylinder	Cone

Shaft Angle	0	Typically 90
Tooth path	Straight	Tapered toward apex

The conical geometry of bevel gears allows for smooth, effective engagement at an angle, making them essential for right angle applications

Straight Vs Spiral Bevel Gears

Spiral bevel gears are for demanding applications. Think high speeds, heavy loads, and quieter operation

Straight bevel gears are perfect for simpler, lower-speed system where cost is a major factor

Feature	Spiral Bevel Gear	Straight Bevel Gear
Operation	Smooth & Quiet	Nosier
Load Capacity	Higher	Higher
Cost	Higher	Lower
Best For	High-speed, Heavy-Load	Low-Speed, Simple Systems

The main difference is the gear teeth engage. Sprial bevel gear have curved teeth that allows them to mesh gradually and smoothly. This gradual engagement minimizes impact and vibration. It’s the reason they operate so quietly, making them ideal for high performance systems.

Straight bevel gears have straight teeth. They engage along the entire tooth face at once. This abrupt contact generates more noise and vibration.

Spiral gears have a higher contact ratio. This means more teeth are in contact at any given moment, distributing the load more effectively.

Criterion	Spiral Bevel Gear	Straight Bevel Gear
Tooth Contact	Gradual, Point Contact	Abrupt, Line Contact
Noise Level	Low	High
Vibration	Minimal	Significant
Manufacturing	Complex	Simpler

Ideal Speed	High RPM	Low to Moderate RPM

Zerol Vs Spiral & Straight Bevel Gears

Feature	Straight Bevel Gear	Spiral Bevel Gear	Zerol Bevel Gear
Spiral Angle	0 °	>0°	0°
Tooth Contact	Abrupt	Gradual	Gradual
Thrust Load	Moderate	High	Moderate

Zerol Vs Hypoid Bevel Gears

Parameter	Zerol Bevel Gear	Hypoid Bevel Gear
Shaft Axes	Intersecting	Non-Intersecting
Spiral Angle	Zero	Non-Zero
Tooth Action	Mostly Rolling	Rolling and Sliding
Key Benefit	Smooth operation, moderate thrust	High torque, quiet, compact design
Common Use	Power tools, machine tools	Automotive differentials, industrial drives

Spiral Vs Zerol Bevel Gears

Parameter	Spiral Bevel Gear	Zerol Bevel Gear
Spiral angle	35° (AGMA / Gleason standard)	0° (curved teeth but no net spiral)
Tooth curvature	Curved	Curved (same generating process as spiral bevel)
Axial thrust	Present, direction depends on hand + rotation direction	Minimal - behaves like straight bevel for thrust purposes
Bidirectional rotation	Not ideal - thrust reverses direction with rotation, complicating bearing design	Well suited - thrust direction stays consistent even when rotation reverses

Load/Torque capacity	Highest	Between straight bevel and spiral bevel
Contact ratio / smoothness	Highest - most gradual tooth engagement	Better than straight bevel, less than spiral
Noise	Quietest	Quieter than straight bevel gear, slightly more than spiral
Bending strength	Good	Comparable or slightly better than straight bevel - larger tooth thickness
Manufacturing	Mainstream	Niche
Sourcing / Tooling availability	Widely avaliable cutters, shops, AGMA data	Hard to find suppliers / shops that cut it accurately
Hand matching requirement	Yes	Yes
Bearing design complexity	Higher - must react thrust in both axial directions on reversal	Lower - simpler bearing arrangement, closer to straight bevel practice
Best for	Continuous, high speed, high load. single direction drives	Oscillating / reversing drivers, or where simpler bearing support is preferred
	High load capacity, but adds thrust reversal complexity on every pitch direction change	Avoids thrust reversal problem inherent to your oscillating duty cycle

Conclusion

Parameter	**Pitch Drive Needs**	**Winner**
Bidirectional Thrust Symmetry	Critical (constant reversing)	Zerol Bevel
Bearing Housing Compactness	Crucial (limited space inside hub)	Zerol Bevel

Shock Load / Bending Resistance	High (wind gusts)	Zerol Bevel
High - Speed Noise Reducion	Low Priority (low operating RPM)	Spiral Bevel

References

Client Challenge

https://www.scribd.com/document/556765185/Gears

Client Challenge

https://www.scribd.com/document/83884983/Bevel-Gear-Calcs

Bevel Gears,Bevel Gear Manufacturers,Bevel Gear,Bevel Gear

Manufacturers, suppliers and exporters of bevel gears, bevel gear manufacturers, bevel gear, bevel gear wholesale, bevel gears manufacturer, bevel gear manufacturer, bevel gears supplier, bevel gear exporter, bevel gear india, plastic bevel gears, bevel gear box, spiral bevel gears, bevel gear drive, bevel gears

http://www.speedreducer.org/bevel-gears.html

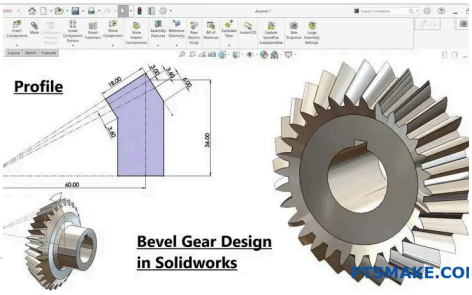
www.kggear.co.jp

https://www.kggear.co.jp/en/wp-content/themes/bizvektor-global-edition/pdf/1.6_Features-of-common-gears_TechnicalData_KGSTOCKGEARS.pdf

The Ultimate Bevel Gear Design Guide

Discover how bevel gears solve intersecting shaft issues with conical design for efficient power transfer, and avoid costly failures with expert insights.

https://www.ptsmake.com/the-ultimate-bevel-gear-design-guide/



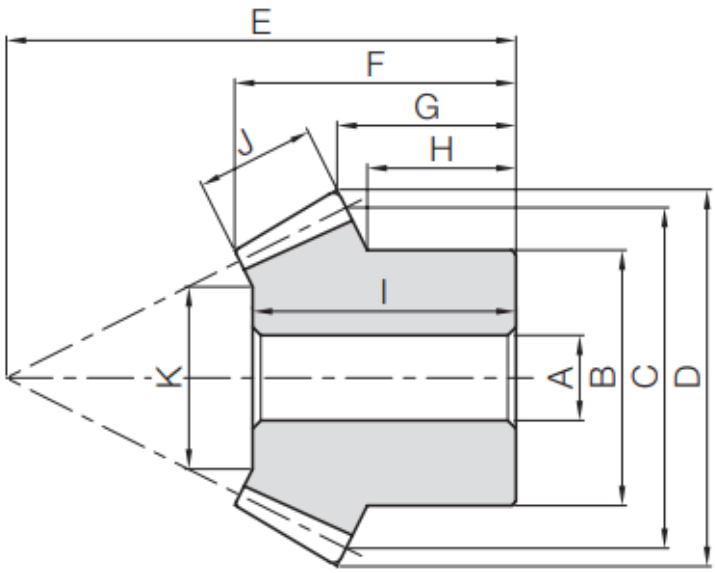
www.cedengineering.com

https://www.cedengineering.com/userfiles/Basic%20Gear%20Fundamentals%20R1.pdf

www.cedengineering.com

https://www.cedengineering.com/userfiles/M06-031%20-%20Basic%20Fundamentals%20of%20Gear%20Drives%20-%20US.pdf

Appendix



- A - Bore
- B - Hub diameter
- C - Pitch diameter
- D - Outside diameter
- E - Mounting distance
- F - Total length
- G - Crown to back
- H - Hub width
- I - Hole length
- J - Face width

- K - Holding surface distance